

ENGR 213 · Applied Ordinary Differential Equations

Lecture 5 Explained: Solutions by Substitutions (Homogeneous, Bernoulli & Linear Arguments)

 **Concordia University** (BCEE Department)

 **Instructor:** Dr. A. Haghighat M.

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 **Textbook:** Zill (7th Ed.), Section 2.5

 **The Transformation Principle:** Substitution is not an independent solution algorithm—it is an **algebraic conversion bridge** that maps difficult non-linear differential equations into elementary **separable** or **linear** ODEs that can be integrated using standard techniques.

[ORIGINAL NON-LINEAR ODE] | ▼ (Pattern Recognition: Structure of RHS)

Bernoulli Equations | 3. Linear Arguments | $M(x,y)dx + N(x,y)dy = 0$ (same degree) | $dy/dx + P(x)y = f(x)y^n$ | $dy/dx = f(Ax + By + C)$
 | → Substitution: $y = ux$ or $x = vx$ | → Substitution: $u = y^{1-n}$ | → Substitution: $u = Ax + By + C$ | → Collapses into: SEPARABLE in u & x | → Collapses into: LINEAR in $u(x)$ | → Collapses into: SEPARABLE in u & x

1. The Introductory Prototype (Lecture 5 Slide 4)

Introductory Prototype · The Pure Ratio ODE

Lecture 5 Slide 4

Equation: Solve $\frac{dy}{dx} = 1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2$

Step-by-Step Derivation:

1. Pattern Recognition: Notice that the right-hand side is exclusively composed of terms in the ratio $\frac{y}{x}$.

2. Substitution: Let $u = \frac{y}{x} \iff y = ux$. By the product rule:

$$\frac{dy}{dx} = u + x \frac{du}{dx}$$

3. Substitute & Simplify:

$$u + x \frac{du}{dx} = 1 + u + u^2 \implies x \frac{du}{dx} = 1 + u^2$$

4. Separate Variables & Integrate:

$$\frac{du}{1+u^2} = \frac{dx}{x} \implies \int \frac{du}{1+u^2} = \int \frac{dx}{x} \implies \arctan(u) = \ln|x| + C$$

$$u(x) = \tan(\ln|x| + C)$$

5. Back-Substitution: Since $u = y/x$:

$$y(x) = x \tan(\ln|x| + C)$$

2. Pattern Recognition: Classification & Substitution Choice

Example 1 · Instant Classification Matrix

Given Differential Equation	Structural Pattern	Classification	Chosen Substitution	Target ODE Form
(a) $\frac{dy}{dx} = (x + y)^2$	Linear combination $(x + y)$	Reduction to Separable	$u = x + y \implies \frac{dy}{dx} = \frac{du}{dx} - 1$	$\frac{du}{dx} = u^2 + 1$ (Separable)
(b) $\frac{dy}{dx} = 2 + e^{2x - y + 1}$	Exponent contains $(2x - y + 1)$	Reduction to Separable	$u = 2x - y + 1 \implies \frac{dy}{dx} = 2 - \frac{du}{dx}$	$\frac{du}{dx} = -e^u$ (Separable)
(c) $\frac{dy}{dx} + 2y = xy^3$	$y' + P(x)y = f(x)y^n$ with $n = 3$	Bernoulli Equation	$u = y^{1-3} = y^{-2}$	$\frac{du}{dx} - 4u = -2x$ (Linear in u)
(d) $\frac{dy}{dx} = y(x + \ln y)$	Rewrite as $\frac{1}{y}y' - \ln y = x$	Logarithmic Linear Form	$v = \ln y \implies v' = \frac{1}{y}y'$	$\frac{dv}{dx} - v = x$ (Linear in v)

3. Homogeneous Equations & Example 2

Core Theory: Homogeneous Functions & Differential Equations

Lecture 5 Slide 8

Definition: A function $f(x, y)$ is homogeneous of degree α if $f(tx, ty) = t^\alpha f(x, y)$.

An equation in differential form $M(x, y)dx + N(x, y)dy = 0$ is homogeneous if both M and N have the same degree α .

- If $N(x, y)$ is algebraically simpler $\implies$ substitute $y = ux \implies dy = u dx + x du$.
- If $M(x, y)$ is algebraically simpler $\implies$ substitute $x = vy \implies dx = v dy + y dv$.

Example 2 · Full Step-by-Step Derivation**Equation:** Solve $(x^2 + y^2)dx + (x^2 - xy)dy = 0$ **Full Step-by-Step Solution:****1. Verify Homogeneity:**

$$M(tx, ty) = (tx)^2 + (ty)^2 = t^2(x^2 + y^2) = t^2 M(x, y) \quad (\alpha = 2)$$

$$N(tx, ty) = (tx)^2 - (tx)(ty) = t^2(x^2 - xy) = t^2 N(x, y) \quad (\alpha = 2)$$

Both coefficients are homogeneous of degree 2.

2. Substitution: Since $N(x, y) = x^2 - xy = x(x - y)$ factors easily, substitute $y = ux \implies dy = u dx + x du$:**3. Substitute into the ODE:**

$$(x^2 + u^2 x^2)dx + (x^2 - x(ux))(u dx + x du) = 0$$

$$x^2(1 + u^2)dx + x^2(1 - u)(u dx + x du) = 0$$

Divide by $x^2 \neq 0$:

$$(1 + u^2)dx + (u - u^2)dx + x(1 - u)du = 0$$

Combine dx terms:

$$[(1 + u^2) + (u - u^2)]dx + x(1 - u)du = 0 \implies (1 + u)dx + x(1 - u)du = 0$$

4. Separate Variables:

$$x(1 - u)du = -(1 + u)dx \implies \frac{dx}{x} + \frac{u - 1}{u + 1}du = 0$$

Rewrite fraction: $\frac{u-1}{u+1} = \frac{(u+1)-2}{u+1} = 1 - \frac{2}{u+1}$

$$\frac{dx}{x} + \left(1 - \frac{2}{u+1}\right)du = 0$$

5. Integrate Both Sides:

$$\ln|x| + u - 2\ln|u+1| = C_1 \implies \ln\left|\frac{x}{(u+1)^2}\right| = -u + C_1$$

$$\frac{(u+1)^2}{|x|} = C_2 e^u \implies (u+1)^2 = Cx e^u$$

6. Back-Substitute $u = \frac{y}{x}$:

$$\left(\frac{y}{x} + 1\right)^2 = Cx e^{y/x} \implies \frac{(x+y)^2}{x^2} = Cx e^{y/x} \implies (x+y)^2 = Cx^3 e^{y/x}$$

(With singular solution $u = -1 \implies y = -x$).

⚠ Concordia Exam Trap: Forgetting the product rule on $dy = u dx + x du$. If you write $dy = u dx$, the du term vanishes and the problem becomes impossible to solve!

4. Bernoulli's Equation & Example 3

Core Theory: Bernoulli Linearization

Lecture 5 Slide 10

Canonical Form: $\frac{dy}{dx} + P(x)y = f(x)y^n \quad (n \neq 0, 1)$

Master Formula: Under the substitution $u = y^{1-n}$, the non-linear equation transforms directly into:

$$\frac{du}{dx} + (1-n)P(x)u = (1-n)f(x)$$

Example 3 · Full Step-by-Step Derivation

Lecture 5 Slide 11

Equation: Solve $x \frac{dy}{dx} + y = x^2 y^2$

Full Step-by-Step Solution:

1. Standardize (Divide by x):

$$\frac{dy}{dx} + \frac{1}{x}y = xy^2 \implies P(x) = \frac{1}{x}, \quad f(x) = x, \quad n = 2$$

2. Substitution: $u = y^{1-n} = y^{1-2} = y^{-1} = \frac{1}{y}$.

3. Transform into Linear ODE: With $(1-n) = 1-2 = -1$:

$$\frac{du}{dx} + (-1)\left(\frac{1}{x}\right)u = (-1)(x) \implies \frac{du}{dx} - \frac{1}{x}u = -x$$

4. Integrating Factor $\mu(x)$:

$$\mu(x) = e^{\int -1/x dx} = e^{-\ln|x|} = \frac{1}{|x|} \implies \mu(x) = \frac{1}{x} \quad (\text{for } x > 0)$$

5. Multiply & Integrate:

$$\begin{aligned} \frac{1}{x} \frac{du}{dx} - \frac{1}{x^2}u &= -1 \implies \frac{d}{dx} \left[\frac{1}{x}u \right] = -1 \\ \frac{1}{x}u &= -x + C \implies u(x) = Cx - x^2 \end{aligned}$$

6. Back-Substitute $u = \frac{1}{y}$:

$$\frac{1}{y} = Cx - x^2 \implies y(x) = \frac{1}{Cx - x^2} = \frac{1}{x(C - x)}$$

(And the singular trivial solution divided out during step 1: $y(x) \equiv 0$)

⚠ Concordia Exam Trap: Failing to divide by x before identifying $P(x)$! Saying $P(x) = 1$ leads to $\mu(x) = e^{-x}$ and ruins the entire integration. Always standardize to $y' + P(x)y = f(x)y^n$ first!

5. Reduction to Separation of Variables & Example 4

Example 4 · Full Step-by-Step Derivation

Lecture 5 Slide 12

Equation: Solve $\frac{dy}{dx} = (-2x + y)^2 - 7$

Full Step-by-Step Solution:

1. Identify Linear Argument: Form is $\frac{dy}{dx} = f(Ax + By + C)$ with $A = -2, B = 1, C = 0$.

2. Substitution: Let $u = -2x + y \implies \frac{du}{dx} = -2 + \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{du}{dx} + 2$.

3. Transform to Separable ODE:

$$\frac{du}{dx} + 2 = u^2 - 7 \implies \frac{du}{dx} = u^2 - 9$$

4. Constant Solutions Pre-Check: $u^2 - 9 = 0 \implies u = \pm 3 \implies y = 2x + 3$ and $y = 2x - 3$.

5. Separate & Integrate via Partial Fractions:

$$\frac{du}{(u-3)(u+3)} = dx \implies \frac{1}{6} \left(\frac{1}{u-3} - \frac{1}{u+3} \right) du = dx$$

$$\ln \left| \frac{u-3}{u+3} \right| = 6x + C_1 \implies \frac{u-3}{u+3} = ce^{6x} \implies u(x) = \frac{3(1+ce^{6x})}{1-ce^{6x}}$$

6. Back-Substitute $u = -2x + y$:

$$-2x + y = \frac{3(1+ce^{6x})}{1-ce^{6x}} \implies y(x) = 2x + \frac{3(1+ce^{6x})}{1-ce^{6x}}$$

(Where $c = 0$ yields $y = 2x + 3$, and $y = 2x - 3$ is a singular lost solution).